MODULE 10

Analytics Projects

Build a Portfolio That Gets You Hired

■ REAL-WORLD PROJECTS

■ LEARNING OUTCOME

By the end of this PDF, you will have a complete framework for executing analytics projects end-to-end, a portfolio of 5 real-world projects across different domains, and the confidence to present your work in interviews.

■ Skills You Will Gain
• Follow the end-to-end analytics project framework
• Complete 5 domain-specific projects with real datasets
• Document projects professionally on GitHub
• Build a data portfolio that stands out
• Answer project-based interview questions
• Know which tools to use for each project type

■ **A portfolio of 3–5 real projects is worth more than any certification. Recruiters want to see what you can DO, not what you know.**

SECTION 1

The Analytics Project Framework

1	**Define the Problem** What business question are you answering? What decision will this enable?
2	**Get the Data** Kaggle, UCI ML Repo, government data, or scrape it yourself
3	**Explore & Clean** EDA + data cleaning — understand what you have before analyzing
4	**Analyze** Apply relevant techniques: aggregation, stats, ML, forecasting
5	**Visualize** Charts and dashboards that communicate the answer clearly
6	**Document Insights** Write 5–7 key findings with supporting evidence
7	**Present** 1-page summary or 5-slide deck — the 'executive report'
8	**Publish** GitHub repo with README + Kaggle notebook or Power BI link

SECTION 2

Project 01 — Sales Performance Analysis

PROJECT 01	Superstore Sales Analysis ■ E-Commerce / Retail
Dataset:	Sample Superstore dataset (Kaggle) — 9994 orders, 21 columns
Steps:	Load & Clean → EDA → Segment Analysis → Profit Deep-dive → Dashboard
Output:	Power BI dashboard + Python EDA notebook + 1-page insight report
Find:	Which category is most profitable? Which region loses money? What drives discounts?

Key Analyses to Perform:

Analysis	Technique	Tool

Monthly revenue trend	Time series line chart	Excel / Power BI
Profit by category & sub-category	Grouped bar chart + heatmap	Python / Power BI
Discount vs Profit correlation	Scatter plot + correlation coefficient	Python
Top 10 products by revenue	Sorted horizontal bar chart	Excel / Python
Shipping mode impact on profit	Box plot comparison	Python

SECTION 3

Project 02 — HR Attrition Analysis

PROJECT 02	Employee Attrition Predictor	■ Human Resources
Dataset:	IBM HR Analytics dataset (Kaggle) — 1470 employees, 35 features	
Steps:	EDA → Identify Risk Factors → Segment by Dept/Role → Visualize → Recommend	
Output:	Attrition dashboard + risk factor report + actionable HR recommendations	
Find:	Who is most likely to leave? What salary band is at risk? Which department has highest churn?	

Key Analyses:

• Attrition rate by Department, Job Role, Age Group, Salary Band • Correlation: OverTime, JobSatisfaction, WorkLifeBalance vs Attrition • Average tenure of employees who left vs stayed • Heat map: Attrition rate by Department × Salary Band • Bar chart: Top 5 factors correlated with attrition

SECTION 4

Project 03 — Financial Data Analysis

PROJECT 03	Personal Finance / Budget Tracker Analysis	■ Finance / Banking
Dataset:	Create your own: 3 months of personal expenses (or use sample bank CSV from Kaggle)	
Steps:	Categorize → Monthly Trends → Spending Heatmap → Budget vs Actual → Forecast	
Output:	Excel dashboard + spending breakdown chart + 3-month forecast	
Find:	Where does most money go? Which category is over budget? Projected savings next month?	

If using stock market data instead:

```
• Download stock prices using yfinance: pip install yfinance • Analyze: daily returns, 30-day moving average, volatility • Compare 3 stocks: RELIANCE, TCS, INFY over 1 year • Visualize: price trend, volume, correlation between stocks
```

SECTION 5

Project 04 — Healthcare Data Analysis

PROJECT 04	Hospital Patient Analysis	■ Healthcare
Dataset:	Heart Disease UCI dataset (Kaggle/UCI) — 303 patients, 14 features	
Steps:	EDA → Risk Factor Analysis → Age/Gender Breakdown → Correlation → Visualize	
Output:	Clinical insights report + risk factor visualization + patient segmentation	
Find:	Which age group is most at risk? Does cholesterol correlate with heart disease? Gender differences?	

Key Analyses:

Question	Chart Type
Age distribution of patients with heart disease	Histogram with disease overlay
Cholesterol vs max heart rate	Scatter plot colored by diagnosis
Disease rate by chest pain type	Grouped bar chart
Correlation of all risk factors	Heatmap
Male vs Female disease rates	Side-by-side bar or pie charts

SECTION 6

Project 05 — Social Media / Marketing Analytics

PROJECT 05	Instagram / YouTube Channel Analytics	■ Marketing / Social Media
Dataset:	Export your own Instagram insights OR use Social Media dataset from Kaggle	
Steps:	Engagement Analysis → Post Performance → Timing Analysis → Content Type → Growth	
Output:	Marketing dashboard + best-performing content report + posting strategy recommendation	
Find:	Best time to post? Which content type gets most engagement? Follower growth trend?	

If using YouTube data:

• YouTube API or Social Blade data • Analyze: views, likes, comments, watch time by video • Find: best-performing video categories, upload timing, title length vs views • Build: Content strategy recommendations based on data

SECTION 7

How to Document Your Project on GitHub

1	**Create Repo** Name it descriptively: 'superstore-sales-analysis' not 'project1'
2	**Write README** Include: Problem Statement, Dataset, Tools Used, Key Findings (with screenshots), How to Run

3	**Add Notebook** Jupyter .ipynb with clean code, markdown cells explaining each section
4	**Screenshots** Add 3–5 chart images to README so visitors see results instantly
5	**Publish** Kaggle Notebooks are public + Power BI publish to web for free dashboard links

README Template Structure:

```
# Project Title ## Problem Statement ## Dataset ## Tools Used ## Key Findings - Finding 1 +
[chart image] - Finding 2 + [chart image] ## How to Run ## Conclusions & Recommendations
```

SECTION 8

Interview Q&A; — Project-Based Questions

Question	How to Answer
Tell me about a data project you did	Use STAR: Situation (business problem) → Task (your goal) → Action (what you did) →
How did you handle missing data in your project?	Name the specific column, state % missing, explain your decision: 'I filled Age with medi
What was the most surprising insight you found?	Pick a counterintuitive finding: 'I expected North to lead, but West had 40% higher profit
How would you present this to a non-technical manager?	Lead with the business impact: 'This analysis shows we're losing ■8L/month from cart a
What would you do differently?	Shows maturity: 'I'd add external data like competitor pricing to better explain the sales c

SECTION 9

Building Your Data Portfolio

Platform	What to Put There
GitHub	All project code, notebooks, README with insights and screenshots
Kaggle	Public notebooks — great for community visibility and competitions
LinkedIn	Post 1 chart + 3-line insight from each project — builds credibility
Tableau Public / Power BI	Published interactive dashboards with shareable links
Personal Website	Portfolio page linking all projects — use GitHub Pages (free)

Portfolio Quality Over Quantity

3 excellent, well-documented projects beat 10 rushed ones. For each project: clean code + clear README + published dashboard + written insights. Recruiters spend 2 minutes on your portfolio — make every second count.

■ PRACTICE QUESTIONS

Q1. What are the 8 steps of the analytics project framework?

Answer: *Define Problem → Get Data → Explore & Clean → Analyze → Visualize → Document Insights → Present → Publish*

Q2. Why is GitHub important for a data analyst portfolio?

Answer: *It shows your code quality, documentation skills, and that your work is real and verifiable — not just resume claims.*

Q3. You found that discount rate has a -0.72 correlation with profit. How do you present this?

Answer: *Lead with business impact: 'Heavy discounting is destroying profit — every 10% discount drops profit margin by 15%. We recommend a discount cap policy.'*

Q4. Name 3 free data sources for analytics projects.

Answer: *Kaggle.com, UCI ML Repository (archive.ics.uci.edu), Google Dataset Search (datasetsearch.research.google.com)*

Q5. What should a project README always include?

Answer: *Problem statement, dataset description, tools used, key findings with chart screenshots, and how to run the code.*

■ BONUS RESOURCES

Kaggle Datasets	kaggle.com/datasets — 50,000+ free datasets across all domains
UCI ML Repo	archive.ics.uci.edu — classic, well-documented datasets
Google Dataset	datasetsearch.research.google.com — search any topic
Data.gov.in	Indian government open data — great for India-specific projects
yfinance (Python)	pip install yfinance — download stock market data instantly
GitHub Pages	Free portfolio website: username.github.io
Kaggle Notebooks	Free GPU + public notebooks — best for portfolio visibility
Interview Prep	stratascratch.com — real interview problems with solutions

■ Analytics Project Cheat Sheet	
Project Framework	Define → Get → Clean → Analyze → Visualize → Present → Publish
README must-haves	Problem + Dataset + Tools + Findings + Screenshots
Project 1	Sales Analysis — Excel/Power BI — Superstore
Project 2	HR Attrition — Python — IBM HR Dataset
Project 3	Finance — Excel — Personal/Bank data
Project 4	Healthcare — Python — Heart Disease UCI
Project 5	Marketing — Python/Power BI — Social data
GitHub naming	project-name-domain (no spaces, lowercase)
Portfolio: 3-5 projects	Quality > Quantity always
Interview: STAR	Situation, Task, Action, Result
Key metric	Always include business impact number
Free hosting	Kaggle + GitHub Pages + Power BI Web